

Antibiotics and risk of subsequent first-time acute myocardial infarction.

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CONTEXT Increasing evidence supports the hypothesis of a causal association between certain bacterial infections and increased risk of developing acute myocardial infarction. If such a causal association exists, subjects who used antibiotics active against the bacteria, regardless of indication, might be at lower risk of developing acute myocardial infarction than nonusers. **OBJECTIVE** To determine whether previous use of antibiotics decreases the risk of developing a first-time acute myocardial infarction. **DESIGN** Population-based case-control analysis. **SETTING** The United Kingdom-based General Practice Research Database comprising 350 general practices. **PATIENTS** A total of 3315 case patients aged 75 years or younger with a diagnosis of first-time acute myocardial infarction between 1992 and 1997 and 13139 controls without myocardial infarction matched to cases for age, sex, general practice attended, and calendar time. **MAIN OUTCOME MEASURES** Use of antibiotics among those who did or did not have a first-time acute myocardial infarction. **RESULTS** Cases were significantly less likely to have used tetracycline antibiotics (adjusted odds ratio [OR], 0.70; 95% confidence interval [CI], 0.55-0.90) or quinolones (adjusted OR, 0.45; 95% CI, 0.21-0.95). No effect was found for previous use of macrolides (primarily erythromycin), sulfonamides, penicillins, or cephalosporins. **CONCLUSIONS** The findings from this large case-control analysis provide further, albeit indirect, evidence for an association between bacterial infections with organisms susceptible to tetracycline or quinolone antibiotics and the risk of acute myocardial infarction. These results of preliminary nature should stimulate more research to further explore the role of infections in the etiology of acute myocardial infarction.